Yacouba BARRO

Statistician-Economist Engineering Student (ISE)

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PROFILE

Statistician-economist engineering student at ISSP (Université Joseph Ki-Zerbo) with a bachelor's degree in economics, specialising in applied econometrics, quantitative macroeconomics and data science. My interests lie in macroeconomic modelling, econometrics applied to West African economies, and the contribution of machine learning to forecasting. Alongside my studies, I build digital tools and online courses that make quantitative methods accessible in Burkina Faso. I am seeking an internship, a scholarship or a doctoral position in these fields.

RESEARCH INTERESTS

Applied macroeconomics and DSGE models • Time-series econometrics • Trade, foreign direct investment and growth in West Africa • Forecasting and machine learning • Development economics and the capability approach

EDUCATION

Since 2023 — **Statistician-Economist Engineer programme (Ingénieur Statisticien Économiste, ISE)**
Institut Supérieur des Sciences de la Population (ISSP), Université Joseph Ki-Zerbo, Ouagadougou
Coursework: econometrics, national accounts (SNA 2008), macroeconomics, applied machine learning. Final-year internship in progress at Coris Assurance Vie.

2023 — **Bachelor's degree (Licence) in Economics, Macroeconomics and Development Management track**
Université Nazi Boni, Bobo-Dioulasso

2018 — **Baccalauréat (secondary school diploma)**
Complexe Scolaire Alpha la Solidarité

RESEARCH AND ACADEMIC PROJECTS

ISSP — **An RBC-type DSGE model for Burkina Faso (2005–2021 data)**
Built and simulated a dynamic stochastic general equilibrium model on Burkinabè data with Dynare and Octave; presented the results in a Beamer (LaTeX) slide deck.

ISSP — **Trade openness, FDI and economic growth in Burkina Faso**
Estimated an ARDL model (short- and long-run relationships) in Python and wrote the econometric report.

ISSP — **Forecasting hourly electricity demand (AEP series, 2004–2018)**
Compared LightGBM, XGBoost, random forests, LSTM (PyTorch) and linear regression. Best model: LightGBM (MAPE = 0.70 %, $R^2 = 0.997$).

EXPERIENCE

Ongoing — **Final-year intern**
Coris Assurance Vie (life insurance), Burkina Faso

Ongoing — **Founder and developer, Faso Carrières**
fasocarrieres.com — education, training and jobs platform for Burkina Faso
- Designed and built the platform (Next.js, React, Node.js, Appwrite, Vercel).
- Wrote online courses on mobile data collection (KoboToolbox, ODK), surveys and sampling, applied econometrics (Stata, R), data analysis (SPSS, Stata), mapping (QGIS), LaTeX, Excel for finance and generative AI.
- Project competing in the 2026 National Digital Innovation Competition (Le Faso Digital).

1 month — **Intern**
Institut National de la Statistique et de la Démographie (INSD), Burkina Faso's national statistics office, Ouagadougou

UJKZ — **Digital assistant**
Université Joseph Ki-Zerbo, Ouagadougou
Updated the data of the campusfaso.com platform.

Freelance	**Graphic designer and video editor** Labels and packaging for cosmetics brands, posters, communication materials and video editing.

DIGITAL AND EDUCATIONAL WORK

Mobile	**FasoPrepas, exam-preparation app** Timed multiple-choice tests, detailed answers and progress tracking for competitive entrance exams (ISE, ASECNA, IFORD, etc.), built with Flutter, Node.js and Appwrite.
Editing	**La RévueCapes** Typeset and edited a preparation guide for the CAPES mathematics teaching exam (LaTeX).

CERTIFICATIONS

Coursera	**Finance & Quantitative Modeling for Analysts (Specialization)** *University of Pennsylvania* coursera.org/verify/specialization/R352N0T1ED2I
Coursera	**Microsoft Power BI Data Analyst** *Microsoft* coursera.org/verify/23P9VXRLDMQJ
Other	**Training in graphic design**

SKILLS

Econometrics	Stata, R, SPSS, Python (scikit-learn, LightGBM, XGBoost, PyTorch), Dynare, Octave
Data	KoboToolbox, ODK, QGIS, Power BI (Power Query), Excel (VBA), SQL, PostgreSQL, Access
Writing	LaTeX and Beamer, Word
Web & mobile	Next.js, React, Node.js, Flutter (Dart), Appwrite, Git and GitHub
Design	Photoshop, Illustrator, InDesign, Canva, CapCut

LANGUAGES

Native	Dioula, Bolon
French	Fluent (working language)
English	Basic; reads technical documentation

COMMUNITY AND INTERESTS

Community	N'Dorola Massakô: designed fundraising posters and communication materials
Interests	Political economy and philosophy of development (Amartya Sen), education, digital innovation, educational content creation
References	Available on request